DDA 3005 — Numerical Methods
Newton's Method: Old and New Stories

Lecture 24

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Repetition

Methods for 1D-Nonlinear Equations

Bisection Method:

- Given an initial interval $[x_\ell, x_r]$ with $f(x_\ell) \cdot f(x_r) < 0$, the bisection method iteratively halves the length of $[x_\ell, x_r]$ to locate a solution x^* with $f(x^*) = 0$.

Fixed-Point Iterations:

- **Idea:** Rewrite $f(x) = 0$ as a fixed-point problem: find x such that $x = g(x)$.
- The fixed-point iteration is given by: $x^{k+1} = g(x^k)$.
- If g is contractive in a neighborhood U of a fixed-point x^* and we choose $x^0 \in U$, then the fixed-point iterates converge linearly to x^*.

Convergence of Fixed-Point Schemes

Lipschitz Continuity and Comments:

- A mapping $g : \mathbb{R} \to \mathbb{R}$ is Lipschitz continuous (contractive) on U with constant $L \geq 0$ iff $|g'(x)| \leq L$ ($L < 1$) for all $x \in U$.

- The convergence result is local: we need to start in a neighborhood of a fixed-point x^* in which g is contractive!
- If $g'(x^*) = 0$ and g is twice continuously differentiable, then the convergence rate is at least quadratic!

Existence:

- If g is contractive on U and $g : U \to U$, i.e., $g(U) \subseteq U$, then g has a unique fixed-point in U.

$\rightsquigarrow$ This result is the celebrated Banach fixed-point theorem.

Newton's Method in 1D

Newton's Method

Newton's method is an iterative method. At each point x^k, we first approximate / linearize f using a first-order Taylor expansion at x^k:

$$f(x) \approx f(x^k) + f'(x^k)(x - x^k)$$

We set the right-hand side to be 0 and solve it:

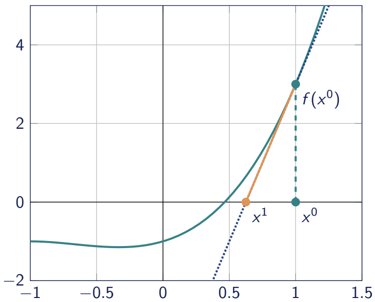
$$x = x^k - \frac{f(x^k)}{f'(x^k)}$$

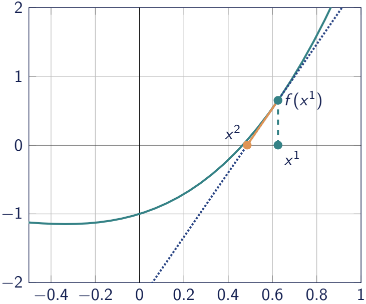
This x is chosen as our next iterate x^{k+1}.

- Here, we assume $f'(x) \neq 0$ at each step!

Newton's Method: Continued

Newton's method approximates the nonlinear function f near x^k by its tangent line $t(x) = f(x^k) + f'(x^k)(x - x^k)$.





Convergence of Newton's Method

Convergence of Newton's Method:

- Newton's method transforms the nonlinear equation $f(x) = 0$ into a fixed-point problem $x = g(x)$ where

$$g(x) = x - \frac{f(x)}{f'(x)}$$

  and hence $g'(x) = f(x)f''(x)/(f'(x))^2$.

- If x^* is a root with $f(x^*) = 0$ and $f'(x^*) \neq 0$, then $g'(x^*) = 0$.
- The convergence of Newton's method in such a case is quadratic:

$$\boxed{|x^{k+1} - x^*| \leq C|x^k - x^*|^2}$$

  for some $C > 0$.

- **But:** x^0 must be sufficiently close to x^* to ensure convergence!

Algorithms for Nonlinear Systems

Fixed-Point Iterations in $\mathbb{R}^n$

The fixed-point problem for $G : \mathbb{R}^n \to \mathbb{R}^n$ is to find a vector $\boldsymbol{x} \in \mathbb{R}^n$ such that

$$\boldsymbol{x} = G(\boldsymbol{x}).$$

The corresponding fixed-point iteration is given by:

$$\boxed{\boldsymbol{x}^{k+1} = G(\boldsymbol{x}^k)}$$

Concepts in $\mathbb{R}^n$:

- Lipschitz continuity can be generalized to n-dimensional mappings.
- G is Lipschitz continuous (on $\mathbb{R}^n$) with constant $L \geq 0$ if

$$\|G(\boldsymbol{x}) - G(\boldsymbol{y})\|_2 \leq L \cdot \|\boldsymbol{x} - \boldsymbol{y}\|_2 \quad \forall\ \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$$

- If $L < 1$, then G is said to be contractive.
- The mapping G is Lipschitz continuous on a convex set $U \subset \mathbb{R}^n$ if and only if $\|DG(\boldsymbol{x})\|_2 \leq L$ for all $\boldsymbol{x} \in U$.

Example and Convergence

The special case $F = \nabla f$:

- Choosing $F = \nabla f$, we want to find critical points of the optimization problem $\min_{\boldsymbol{x}} f(\boldsymbol{x})$.
- The gradient mapping ∇f is Lipschitz continuous (with constant L) iff the Hessian $\nabla^2 f$ is bounded ($\|\nabla^2 f(\boldsymbol{x})\| \leq L$ for all $\boldsymbol{x}$).
- Typical choice of G: $G(\boldsymbol{x}) = \boldsymbol{x} - \alpha \nabla f(\boldsymbol{x})$ where $\alpha > 0$ is a step size.

Convergence:

- Conditions for convergence are similar to the 1D case.
- If G is contractive on U, then the fixed-point scheme $\boldsymbol{x}^{k+1} = G(\boldsymbol{x}^k)$ converges to a fixed-point $\boldsymbol{x}^* \in U$ as long as $\boldsymbol{x}^0 \in U$.
- The convergence is typically linear with rate $\eta \approx \|DG(\boldsymbol{x}^*)\|$.
- If $DG(\boldsymbol{x}^*) = \mathbf{0}$, then the convergence is quadratic.

Example: Lipschitz Continuity

Consider the function

$$f(\boldsymbol{x}) = x_1^4 + x_1^2 + x_2^2 + 2x_1x_2 + \frac{1}{4}.$$

Is the gradient mapping ∇f Lipschitz continuous on $\mathbb{R}^2$?

Newton's Method

In n dimensions, Newton's method has the form

$$\boxed{\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - DF(\boldsymbol{x}^k)^{-1}F(\boldsymbol{x}^k)}$$

where $DF(\boldsymbol{x})$ is the Jacobian matrix of F at $\boldsymbol{x}$.

- In practice, we do not explicitly invert the matrix $DF(\boldsymbol{x}^k)$ but instead solve the linear system

$$DF(\boldsymbol{x}^k)\boldsymbol{d}^k = -F(\boldsymbol{x}^k)$$

to obtain the Newton direction $\boldsymbol{d}^k$. The next iterate is then given as

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \boldsymbol{d}^k.$$

Example: Newton's Method

We consider the nonlinear system

$$F(\boldsymbol{x}) = \boldsymbol{0} \quad \text{with} \quad F_1(\boldsymbol{x}) = x_1^2 + x_2^2 - 1, \;\; F_2(\boldsymbol{x}) = x_1^4 - x_2^4 + x_1 x_2.$$

The Jacobian of F is given by

$$DF(\boldsymbol{x}) = \begin{bmatrix} 2x_1 & 2x_2 \\ 4x_1^3 + x_2 & -4x_2^3 + x_1 \end{bmatrix}.$$

For the initial point $\boldsymbol{x}^0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^\top$, we then obtain:

$$\boldsymbol{x}^1 = \boldsymbol{x}^0 - DF(\boldsymbol{x}^0)^{-1} F(\boldsymbol{x}^0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 & 2 \\ 5 & -3 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 11 \\ 13 \end{bmatrix}.$$

`iter`	$(\boldsymbol{x}^k)^\top$	$\lVert F(\boldsymbol{x}^k) \rVert$	$\lVert \boldsymbol{x}^k - \boldsymbol{x}^* \rVert$
2	$[0.548, 0.848]$	$4.29 \cdot 10^{-2}$	$2.28 \cdot 10^{-2}$
3	$[0.526, 0.851]$	$9.37 \cdot 10^{-4}$	$5.18 \cdot 10^{-4}$
4	$[0.526, 0.851]$	$5.03 \cdot 10^{-7}$	$2.78 \cdot 10^{-7}$
5	$[0.526, 0.851]$	$1.46 \cdot 10^{-13}$	$8.02 \cdot 10^{-14}$

Convergence of Newton's Method in $\mathbb{R}^n$

Convergence:

- The convergence rate of Newton's method for nonlinear system is generally quadratic provided that the Jacobian matrix $DF(\boldsymbol{x}^*)$ is nonsingular.
- The method must be started sufficiently close to a solution to ensure convergence.

Costs of Newton's Method:

- Computing the Jacobian matrix costs n^2 scalar function evaluations.
- Solving the linear system requires $\mathcal{O}(n^3)$ computational operations.

A Globalized Newton Method

A Globalized Newton Method

Current Disadvantage:

- Newton's method (and other fixed-point schemes) may fail to converge when started far away from a solution.

Discussion and Idea:

- Similar to Newton's method for optimization problems, we can use damped Newton steps of the form:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$$

  where $\boldsymbol{d}^k$ is the Newton direction and α_k is a suitable step size to ensure progress towards a solution.

- The step size α_k reduces the Newton step when it is too large.
- In optimization problems, α_k is chosen to ensure decrease of the objective function f: $f(\boldsymbol{x}^{k+1}) < f(\boldsymbol{x}^k)$.

$\rightsquigarrow$ How should we select the step size for nonlinear equations $F(\boldsymbol{x}) = \mathbf{0}$?

Globalization: Merit Function

To globalize Newton's method for nonlinear equations, we consider the merit function problem:

$$\min_{\boldsymbol{x}} \; m(\boldsymbol{x}) := \frac{1}{2}\|F(\boldsymbol{x})\|^2. \tag{1}$$

Observations:

- $\boldsymbol{x}^*$ solves $F(\boldsymbol{x}^*) = \mathbf{0} \iff m(\boldsymbol{x}^*) = 0$, i.e., $\boldsymbol{x}^*$ is a global solution of problem (1).
- It holds that $\nabla m(\boldsymbol{x}) = DF(\boldsymbol{x})^\top F(\boldsymbol{x})$.

$\rightsquigarrow$ If $DF(\boldsymbol{x})$ is invertible (for all $\boldsymbol{x}$), then stationary points of m need to satisfy $F(\boldsymbol{x}) = \mathbf{0}$.

- Let $\boldsymbol{d}$ be the Newton direction with $DF(\boldsymbol{x})\boldsymbol{d} = -F(\boldsymbol{x})$, then:

$$\nabla m(\boldsymbol{x})^\top \boldsymbol{d} = F(\boldsymbol{x})^\top DF(\boldsymbol{x})\boldsymbol{d} = -\|F(\boldsymbol{x})\|^2 < 0.$$

$\rightsquigarrow$ The Newton direction is always a descent direction for m!

The Globalized Newton Method

The Globalized Newton Method

1. Initialization: Choose an initial point $\boldsymbol{x}^0 \in \mathbb{R}^n$.

For $k = 0, 1, 2, \ldots$:

2. STOP, if $\|\nabla m(\boldsymbol{x}^k)\| \leq \texttt{tol}$.
3. Compute $\boldsymbol{d}^k$ by solving the Newton equation $DF(\boldsymbol{x}^k)\boldsymbol{d}^k = -F(\boldsymbol{x}^k)$.
4. Perform backtracking on the merit function m to calculate a step size α_k. Set $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.

Comments:

- This is a full analog of the globalized Newton method for optimization problems: $m \rightsquigarrow f$.
- Locally, under certain assumptions, the algorithm turns into a pure Newton method with $\alpha_k = 1$ for all k sufficiently large.

$\rightsquigarrow$ Fast local (quadratic) convergence of $\{\boldsymbol{x}^k\}_k$ can be observed.

Application: Constrained Optimization & KKT Conditions

Equality Constrained Optimization

We consider and discuss nonlinear programs with equality constraints:

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad h(\boldsymbol{x}) = \mathbf{0}.$$

where $f:\mathbb{R}^n \to \mathbb{R}$ and $h:\mathbb{R}^n \to \mathbb{R}^p$ are given.

Let $\boldsymbol{x}^*$ be a local solution of this problem. Then, according to the KKT-conditions there exists $\boldsymbol{\mu}^* \in \mathbb{R}^p$:

$$\nabla_x L(\boldsymbol{x}^*, \boldsymbol{\mu}^*) = \nabla f(\boldsymbol{x}^*) + \nabla h(\boldsymbol{x}^*)\boldsymbol{\mu}^* = \mathbf{0} \quad \text{and} \quad h(\boldsymbol{x}^*) = \mathbf{0}.$$

- The KKT-conditions define a nonlinear system of equations with $n+p$ unknowns and $n+p$ equations.

$\rightsquigarrow$ We can apply Newton's method to determine a solution of the nonlinear equation:

$$F(\boldsymbol{x}, \boldsymbol{\mu}) := \begin{bmatrix} \nabla_x L(\boldsymbol{x}, \boldsymbol{\mu}) \\ h(\boldsymbol{x}) \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}.$$

Generating Lagrange-Newton Steps

Let $(\boldsymbol{x}^k, \boldsymbol{\mu}^k)$ be the current iterate. Then, Newton's step $\boldsymbol{d}^k$ for the equation $F(\boldsymbol{x}, \boldsymbol{\mu}) = 0$ is given by

$$DF(\boldsymbol{x}^k, \boldsymbol{\mu}^k)\boldsymbol{d}^k = -F(\boldsymbol{x}^k, \boldsymbol{\mu}^k).$$

In particular, we can calculate

$$DF(\boldsymbol{x}, \boldsymbol{\mu}) = \begin{pmatrix} \nabla^2_{xx} L(\boldsymbol{x}, \boldsymbol{\mu}) & \nabla^2_{\mu x} L(\boldsymbol{x}, \boldsymbol{\mu}) \\ \nabla h(\boldsymbol{x})^\top & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \nabla^2_{xx} L(\boldsymbol{x}, \boldsymbol{\mu}) & \nabla h(\boldsymbol{x}) \\ \nabla h(\boldsymbol{x})^\top & \mathbf{0} \end{pmatrix}$$

and the complete Newton update simplifies to:

$$\begin{pmatrix} \nabla^2_{xx} L(\boldsymbol{x}^k, \boldsymbol{\mu}^k) & \nabla h(\boldsymbol{x}^k) \\ \nabla h(\boldsymbol{x}^k)^\top & \mathbf{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{d}^k_x \\ \boldsymbol{d}^k_\mu \end{pmatrix} = \begin{pmatrix} -\nabla_x L(\boldsymbol{x}^k, \boldsymbol{\mu}^k) \\ -h(\boldsymbol{x}^k) \end{pmatrix}$$

and $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k_x$, $\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \alpha_k \boldsymbol{d}^k_\mu$.

$\rightsquigarrow$ This special version is known as the Lagrange-Newton method.

Numerical Experiment

We consider the problem

$$\min_{\boldsymbol{x}} \; f(\boldsymbol{x}) = (x_1^2 - 1)(x_1^2 - 9) + 8x_1x_2 + (x_2^2 - 1)(x_2^2 - 9)$$

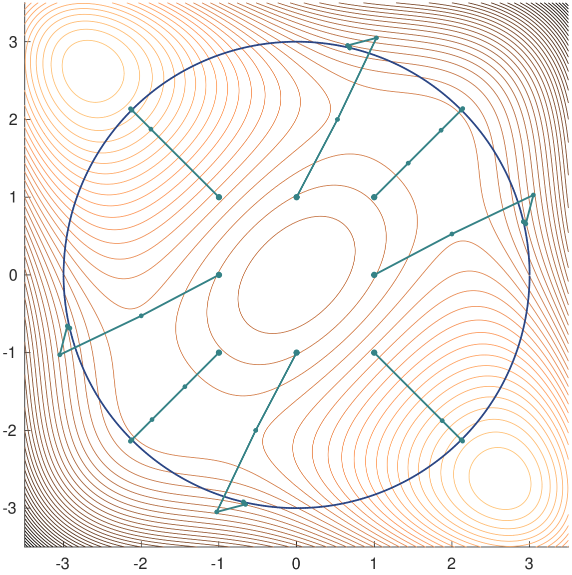
subject to $\|\boldsymbol{x}\| = 3$. Setting $h(\boldsymbol{x}) := \frac{1}{2}\|\boldsymbol{x}\|^2 - \frac{9}{2}$, we then have

$$\nabla_x L(\boldsymbol{x}, \mu) = \begin{bmatrix} 4x_1(x_1^2 - 5) + 8x_2 \\ 4x_2(x_2^2 - 5) + 8x_1 \end{bmatrix} + \mu \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Remarks:

- We implement a standard backtracking strategy for m.
- We test eight different initial points with $x_i^0 \in \{-1, 0, 1\}$ and $\mu = 1$.
- Due to the small size of the problem, we can use `MATLAB`'s backslash or `linalg.solve` to solve the Newton systems.
- Newton's method converges quadratically within $6-7$ steps.

Solution Paths



Inexact Newton Steps

Inexact Newton Steps

It is also possible to solve the linear system

$$DF(\boldsymbol{x}^k) \cdot \boldsymbol{d}^k = -F(\boldsymbol{x}^k)$$

only approximately up to a certain accuracy and to generate inexact and cheaper Newton steps.

Strategy: Find an approximate direction $\boldsymbol{d}^k$ such that:

$$\|DF(\boldsymbol{x}^k)\boldsymbol{d}^k + F(\boldsymbol{x}^k)\| \leq \texttt{tol}, \quad \texttt{tol} \geq 0.$$

Observations:

- If $DF(\boldsymbol{x}^k)$ is symmetric, we can generally apply the CG-method to obtain such an approximate direction.

$\rightsquigarrow$ This is especially effective in the optimization case when $F = \nabla f$ and $DF = \nabla^2 f$ and when n is large!

Inexact Newton Steps: Discussion

Why does this work? ...

- Adaptive choices of `tol`, i.e.,

$$\texttt{tol} \equiv \texttt{tol}_k = \rho_k \|F(\boldsymbol{x}^k)\|, \quad \rho_k \to 0$$

  allow to ensure fast local convergence!

- **Interpretation**: Closer to a solution / stationary point of the problem, we compute the Newton step with higher accuracy.

$\rightsquigarrow$ There are many more techniques and options to boost the performance of Newton's method in large-scale settings!

Questions?